



GIRRAWEE HIGH SCHOOL

2007

**TRIAL HIGHER SCHOOL CERTIFICATE
EXAMINATION**

Mathematics Extension 2

General Instructions

- Reading time – 5 minutes
- Working time – 3 hours
- Write using black or blue pen
- Board-approved calculators may be used
- A table of standard integrals is provided at the back of this paper
- All necessary working should be shown in every question
- Marks may be deducted for careless or badly arranged work

Total marks – 120

- Attempt Questions 1 – 8
- All questions are of equal value
- Start a separate piece of paper for each question.
- Put your name and the question number at the top of each sheet.

Total marks – 120

Attempt Questions 1 – 8

All questions are of equal value

Answer each question on a SEPARATE piece of paper clearly marked Question 1, Question 2, etc. Each piece of paper must show your name.

Question 1 (15 marks) Use a *separate* piece of paper **Marks**

- a) Find $\int \frac{\sin \theta d\theta}{\cos^5 \theta}$ 2
- b) By completing the square, evaluate $\int_{-1}^0 \frac{dx}{\sqrt{1-2x-x^2}}$ 3
- c) (i) Find A, B and C if $\frac{16}{(x^2+4)(2-x)} = \frac{Ax+B}{(x^2+4)} + \frac{C}{(2-x)}$ 2
- (ii) Hence find $\int \frac{16}{(x^2+4)(2-x)} dx$ 2
- d) Use the substitution $x = 3 \sin \theta$ to find $\int \frac{x^2}{\sqrt{9-x^2}} dx$ 3
- e) Using $t = \tan \frac{x}{2}$ find $\int \frac{dx}{1+\sin x}$ 3

Question 2 (15 marks) Use a *separate* piece of paper

- a) Let $z = 2 - i$ and $\omega = 1 + 3i$, find;
- (i) $z\bar{\omega}$ 1
- (ii) $\frac{2}{z}$ 1
- b) Let $\alpha = \frac{4i}{-1+i\sqrt{3}}$
- (i) Express α in modulus argument form 2
- (ii) Express α^5 in modulus argument form 2
- (iii) Hence express α^5 in the form $x + iy$ 1
- c) Sketch the region on the Argand diagram where the inequalities 3
- $$\frac{\pi}{4} \leq \arg(z-i) \leq \frac{3\pi}{4} \quad \text{and} \quad |z-i| \leq 2$$
- hold simultaneously

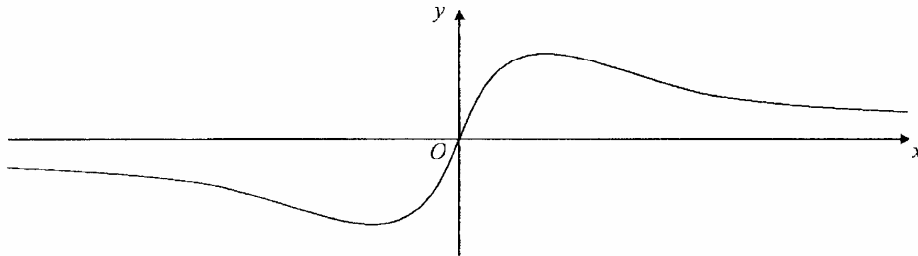
Question 2 (continued)**Marks**

- d) The point A represents the real number $z_1 = 1$. The point B represents the complex number $z_2 = (1 + \sqrt{3}) + i$. If $ABCD$ is a square in anti-clockwise rotational order, find;
- (i) the complex number z_3 represented by the point D . 2
 - (ii) the complex number z_4 represented by the point C . 2
 - (iii) the complex number z_5 represented by the point of intersection of the diagonals AC and BD . 1

Question 3 (15 marks) Use a *separate* piece of paper

- a) Sketch $y = \frac{3x^2}{4 - x^2}$ showing all asymptotes. 3

- b) The diagram shows the graph of $y = f(x)$ which has range $-1 \leq y \leq 1$



Draw separate one-third page sketches of the graphs of the following;

- (i) $y = |f(x)|$ 1
 - (ii) $y = f(x+1)$ 1
 - (iii) $y = [f(x)]^2$ 2
 - (iv) $y = \frac{1}{f(x)}$ 2
 - (v) $y = e^{f(x)}$ 2
- c) It is known that $P(x) = x^4 - 4x^3 + 5x^2 + ax + b$ is divisible by $(x-2)^2$
- (i) Find the values of a and b . 3
 - (ii) Factorise $P(x)$ into linear factors 1

Question 4 (15 marks) Use a *separate* piece of paper

Marks

- a) A curve has the equation $x^3 + 2xy - 4y^2 = 10$. 2
Find an expression for $\frac{dy}{dx}$ as a function of x and y .
- b) A cylindrical hole of radius 1 cm is bored through the centre of a sphere of radius 3 cm. Using the method of cylindrical shells, calculate the exact volume of the sphere that remains. 4
- c) For the conic $\frac{x^2}{25} + \frac{y^2}{16-\lambda} = 1$ find ;
(i) the values of λ that makes the conic an ellipse with the foci on the x axis. 2
(ii) the values of λ that makes the conic a rectangular hyperbola. 1
(iii) the coordinates of the foci and the equations of the directrices when $\lambda = 0$ 2
- d) How many different ways are there of seating four married couples at a circular table with men and women in alternate positions and no wife next to her husband? 4

Question 5 (15 marks) Use a *separate* piece of paper

- a) Suppose α, β and γ are the three roots of the polynomial equation $x^3 + x + 12 = 0$
(i) Find $\alpha^2 + \beta^2 + \gamma^2$ 2
(ii) Hence explain why only one of the roots is real 1
(iii) The real root is denoted by α . Prove that $-3 < \alpha < -2$ 1
(iv) Hence prove that the modulus of each of the other roots lies between 2 and $\sqrt{6}$ 3
- b) A woman of mass M kg jumps vertically (feet first) from a rock ledge into a river below.
When she is falling at v m/s, she encounters air resistance equal to $\frac{Mv}{10}$ Newtons.
She hits the water at a speed of V m/s.
Let x be the displacement below the rock ledge at time t seconds after jumping.
(i) Show that $\ddot{x} = g - \frac{v}{10}$, where g is the acceleration due to gravity 1
(ii) If it takes one second for her feet to hit the water, using $g = 10 \text{ m/s}^2$ show that; 3
$$V = 100 \left(1 - e^{-\frac{1}{10}} \right)$$

(iii) Find the height of the rock ledge above the water, to the nearest 0.1 metre 4

Question 6 (15 marks) Use a separate piece of paper

Marks

a) By considering the series $1 + t + t^2 + t^3 + \dots + t^n$, or otherwise;

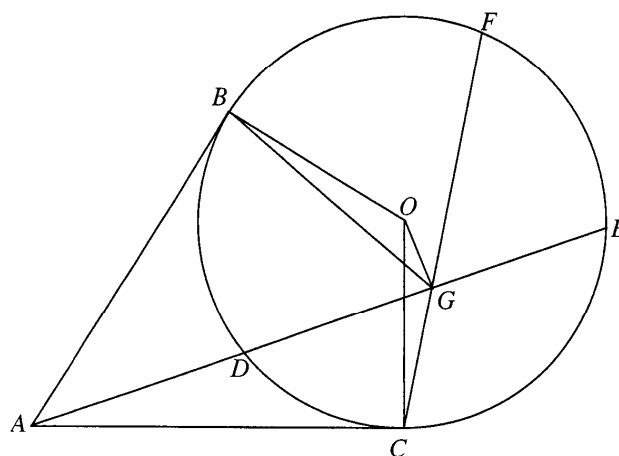
(i) sum the series $1 + 2t + 3t^2 + 4t^3 + \dots + nt^{n-1}$

2

(ii) hence evaluate $1 + 2 \times 2 + 3 \times 2^2 + 4 \times 2^3 + \dots + 20 \times 2^{19}$

1

b)



In the diagram, AB and CA are the tangents from A to the circle with centre O , meeting the circle at B and C .

ADE is a secant of the circle. G is the midpoint of DE . CG produced meets the circle at F .

(Note: on page 10 there is a copy of the above diagram that you can use, please detach and include with your solutions)

(i) Show that $ABOC$ is a cyclic quadrilateral

2

(ii) Show that $AOGC$ is a cyclic quadrilateral

2

(iii) Hence prove that $BF \parallel AE$

3

Question 6 (continued)

Marks

- c) The roof of a sports stadium has an elliptical base with a major axis of length $2a$ and minor axis of length $2b$. The two identical sloping tops are inclined at 30° to the base.

$PQRS$ represents a rectangular cross-section of thickness δx taken x units from the centre O of the ellipse.

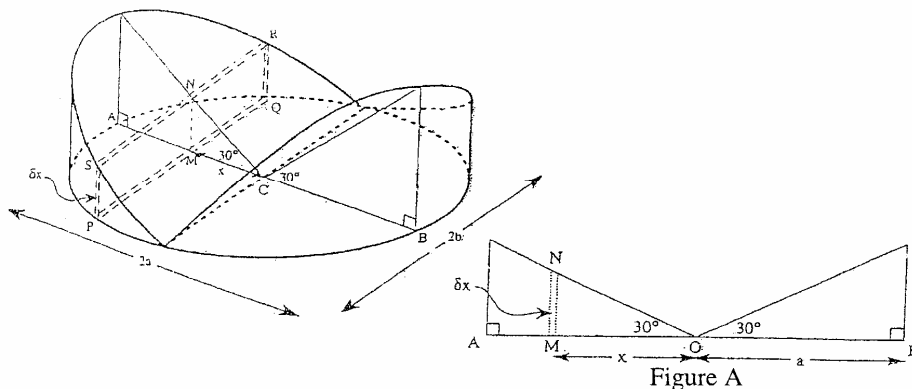


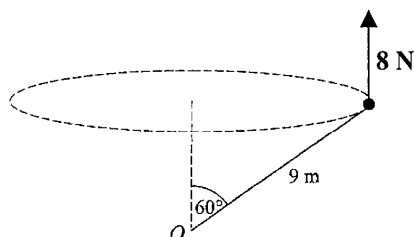
Figure A shows a side view of the stadium if sliced in half along AB

- | | |
|---|---|
| (i) Find the height MN of the rectangular cross-section $PQRS$ | 1 |
| (ii) Show that the area A of the rectangle $PQRS$ is given by $A = \frac{2b}{a\sqrt{3}} x \sqrt{a^2 - x^2}$ | 2 |
| (iii) Calculate the volume of the stadium roof. | 2 |

Question 7 (15 marks) Use a *separate* piece of paper

Marks

a)



A toy aircraft of mass 0.5 kg is attached to one end of a string of length 9 m . The other end of the string is attached to a fixed point O and moves with constant angular velocity.

The string is taut, and makes an angle of 60° with the upward vertical at O .

In a simplified model of the motion, the aircraft is treated as a particle and the force of the air on the aircraft is taken to act vertically upwards with magnitude 8 Newtons . (Use $g = 10 \text{ m/s}^2$)

- | | |
|--|---|
| (i) Resolve the forces on the aircraft in the horizontal and vertical directions | 2 |
| (ii) Find the tension in the string | 1 |
| (ii) Find the speed of the aircraft in m/s . | 2 |

b) $P\left(cp, \frac{c}{p}\right)$ is a point on the hyperbola $xy = c^2$

- | | |
|---|---|
| (i) Show that the equation of the tangent at P has the equation $x + p^2y = 2cp$ | 2 |
| (ii) The tangent at P cuts the x axis and y axis at A and B respectively. Find the coordinates of A and B . | 1 |
| (iii) Q is the fourth vertex of the rectangle $OAQB$. Show that the locus of Q is another rectangular hyperbola. | 2 |

c) (i) Let $I_n = \int \tan^n \theta d\theta$, show that $I_n = \frac{1}{n-1} \tan^{n-1} \theta - I_{n-2}$ 3

(ii) Hence find the exact value of $\int_0^{\frac{\pi}{4}} \tan^4 \theta d\theta$ 2

Question 8 (15 marks) Use a *separate* piece of paper

Marks

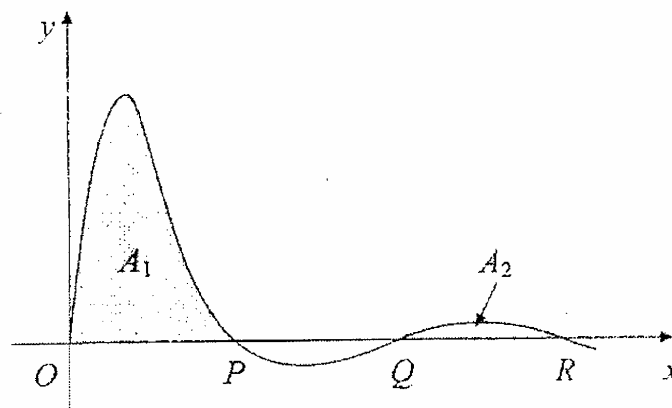
a) (i) Given that $y = x \sin x$, find $\frac{d^2 y}{dx^2}$

1

(ii) Prove by induction that $\frac{d^{2n} y}{dx^{2n}} = (-1)^n (x \sin x - 2n \cos x)$

3

b)



The diagram shows a sketch of part of the curve C with equation

$$y = e^{-x} \sin x, \quad x \geq 0$$

(i) Find the coordinates of the points P , Q and R where C cuts the x axis.

1

(ii) Use integration by parts to show that;

3

$$\int e^{-x} \sin x \, dx = -\frac{1}{2} e^{-x} (\sin x + \cos x) + c$$

(iii) The terms $A_1, A_2, \dots, A_n$ represent areas between C and the x axis for successive portions of C where y is positive. The areas represented by A_1 and A_2 are shown in the diagram.

2

Show that $A_n = \frac{1}{2} (e^{(1-2n)\pi} + e^{(2-2n)\pi})$

(iv) Show that $A_1 + A_2 + A_3 + \dots$ is a geometric series and that $S_\infty = \frac{e^\pi}{2(e^\pi - 1)}$

3

(v) Given that $\int_0^\infty e^{-x} \sin x \, dx = \frac{1}{2}$, find the exact value of

2

$$\int_0^\infty |e^{-x} \sin x| \, dx$$

Extension 2 Trial Solutions 2007

Question 15

a) $\int \frac{\sin \theta d\theta}{\cos^2 \theta}$
 $u = \cos \theta$
 $du = -\sin \theta d\theta$
 $= - \int \frac{du}{u^2}$
 $= \frac{1}{u} + c$
 $= \frac{1}{\cos \theta} + c$

b) $\int \frac{dx}{\sqrt{1-2x-x^2}}$
 $= \int \frac{dx}{\sqrt{2-(x+1)^2}}$
 $= \int \frac{dx}{\sqrt{2-(\frac{x+1}{\sqrt{2}})^2}}$
 $= \sin^{-1}(\frac{x+1}{\sqrt{2}}) + c$
 $= \sin^{-1}(\frac{x}{\sqrt{2}} + \frac{1}{\sqrt{2}}) + c$
 $= \frac{\pi}{4}$

c) a) $A(x+B)(2-x) + C(x^2+4) = 16$
 $x=2$
 $8C = 16$
 $C = 2$
 $2A+B=0$
 $2A-B=4$
 $A=2, B=-4, C=2$
 $\int \frac{16}{(x^2+4)(2-x)} dx = \int \left[\frac{2x}{x^2+4} + \frac{2}{x^2+4} + \frac{2}{2-x} \right] dx$
 $= \log(x^2+4) + 2 \tan^{-1} \frac{x}{2} - 2 \log(2-x) + c$
 $= \log \left[\frac{x^2+4}{(2-x)^2} \right] + 2 \tan^{-1} \frac{x}{2} + c$

d) $\int \frac{x^2}{\sqrt{9-x^2}} dx$
 $x = 3 \sin \theta$
 $dx = 3 \cos \theta d\theta$
 $= \int \frac{9 \sin^2 \theta \cdot 3 \cos \theta d\theta}{3 \cos \theta}$
 $= \int 9 \sin^2 \theta d\theta$
 $= \frac{9}{2} \int (1 - \cos 2\theta) d\theta$
 $= \frac{9}{2} \theta - \frac{9}{4} \sin 2\theta + c$
 $= \frac{9}{2} \sin^{-1} \frac{x}{3} - \frac{9}{4} \cdot \frac{x}{3} \cdot \frac{\sqrt{9-x^2}}{3} + c$
 $= \frac{9}{2} \sin^{-1} \frac{x}{3} - \frac{1}{2} x \sqrt{9-x^2} + c$

$x = \tan \frac{\pi}{2}$
 $dx = \frac{2dt}{1+t^2}$

e) $\int \frac{dx}{1+\sin x}$
 $= \int \frac{2dt}{1+t^2}$
 $= \int \frac{2 dt}{1+t^2}$
 $= \int \frac{2 dt}{(t+1)^2 + 1}$
 $= -\frac{2}{t+1} + c$
 $= -\frac{2}{\tan \frac{x}{2} + 1} + c$

Question 2 (15)

a) (i) $zw = (2-2i)(1-3i)$
 $= 2-6i-2i-3$
 $= -1-7i$

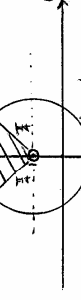
(ii) $\frac{z}{w} = \frac{2i}{1-3i}$
 $= \frac{2i(1+3i)}{(1-3i)(1+3i)}$
 $= \frac{2i(1+3i)}{1-9i^2}$
 $= \frac{2i(1+3i)}{10}$
 $= \frac{2i(1+3i)}{5}$

b) $\alpha = \frac{4i}{-1+2i\sqrt{3}}$
 $= \frac{4i(-1-2i\sqrt{3})}{(-1+2i\sqrt{3})(-1-2i\sqrt{3})}$
 $= \frac{4i(-1-2i\sqrt{3})}{1-12i^2}$
 $= \frac{4i(-1-2i\sqrt{3})}{13}$
 $= \frac{4i(-1-2i\sqrt{3})}{13}$

(ii) $\alpha^5 = 32 \left(\cos \frac{5\pi}{6} + i \sin \frac{5\pi}{6} \right)$
 $= 32 \left(-\frac{\sqrt{3}}{2} - \frac{1}{2}i \right)$
 $= -16\sqrt{3} - 16i$
 $\arg(z) = \frac{5\pi}{6}$
 $\arg(z^5) = \frac{5\pi}{6}$

(iii) $\alpha^5 = 32 \left(\cos \frac{5\pi}{6} + i \sin \frac{5\pi}{6} \right)$
 $= 32 \left(-\frac{\sqrt{3}}{2} - \frac{1}{2}i \right)$
 $= -16\sqrt{3} - 16i$
 $\arg(z) = \frac{5\pi}{6}$
 $\arg(z^5) = \frac{5\pi}{6}$

c) $|z-1|=2$



d) (z_3)



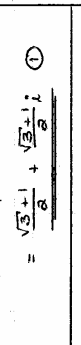
(i) $z_3 - z_1 = i(z_2 - z_1)$
 $z_3 - 1 = i(\sqrt{3}i + 1)$
 $z_3 = \sqrt{3}i - 1$

(ii) $z_4 = z_2 + z_3 - z_1$
 $= (1+i\sqrt{3}) + i + \sqrt{3}i - 1$
 $= \sqrt{3}i + (1+i\sqrt{3})i$

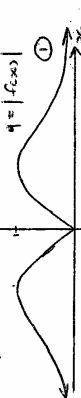
(iii) $z_5 = \text{midpoint of AC}$
 $= \frac{\sqrt{3} + 1 + \sqrt{3}i + 1}{2}$

Question 3 (15)

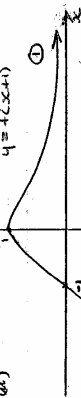
a) $y = \frac{3x^2}{4-x^2}$
 $= -3 + \frac{12}{4-x^2}$
 $\therefore \text{HA: } y = -3$
 $\text{VA: } x = \pm 2$



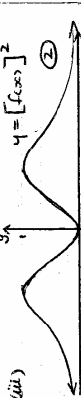
b) (i) $y = |f(x)|$



(ii) $y = f(x+1)$



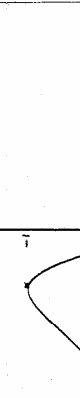
(iii) $y = [f(x)]^2$



(iv) $y = \frac{1}{f(x)}$



(v) $y = e^{f(x)}$



c) (i) $P(x) = x^4 - 4x^3 + 5x^2 + 10x + 6$
 $P'(x) = 4x^3 - 12x^2 + 10x + 6$

$P(2) = 0$

$16 - 32 + 20 + 20 + 6 = 0$
 $8a + b = -4$

$P'(2) = 0$

$32 - 48 + 20 + a = 0$
 $a = -4, b = 4$

(ii) $P(x) = x^4 - 4x^3 + 5x^2 - 4x + 4$
 $= (x^2 - 2x + 2)(x^2 - 2x + 2)$

Question 4 (15)

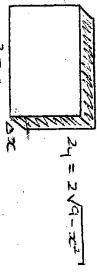
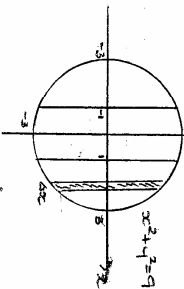
a) $3x^3 + 2xy - 4y^2 = 10$

$3x^2 + (2x) \frac{dy}{dx} + 4y \frac{dy}{dx} - 8y \frac{dy}{dx} = 0$

$(2x - 8y) \frac{dy}{dx} = -3x^2 - 2y$

$\frac{dy}{dx} = \frac{3x^2 + 2y}{8y - 2x}$

b)



$A(x) = 4\pi x \sqrt{9-x^2}$

$\Delta V = 4\pi x \sqrt{9-x^2} \Delta x$

$V = \lim_{x \rightarrow 0} \sum_{i=1}^n 4\pi x_i \sqrt{9-x_i^2} \Delta x$

$= 2\pi \int_{-3}^3 2x \sqrt{9-x^2} dx$

$= -2\pi \left[\frac{2}{3} (9-x^2)^{3/2} \right]_{-3}^3$

$= -\frac{4\pi}{3} (0 - 8\sqrt{8})$
 $= \frac{64\sqrt{2}\pi}{3}$

c) $0 < 16 - \lambda < 25$
 $-16 < -\lambda < 9$
 $-9 < \lambda < 16$

$16 - \lambda = -25$

$\lambda = 41$

$\frac{x^2}{25} + \frac{y^2}{16} = 1$

$a^2 = 25, b^2 = 16$
 $a = 5, b = 4$

$1 - e^2 = \frac{16}{25}$
 $e^2 = \frac{9}{25}$
 $e = \frac{3}{5}$

$\therefore \text{foci: } (\pm 3, 0) \text{ directrices: } x = \pm \frac{25}{3}$

d) Place men first = 3!

With A has 2 spots
 B has 1 spot for each of the other
 women

ways = $3! \times 2 \times 1 \times 1 = 12$

$\frac{1}{12}$

$\frac{1}{12}$

Question 5 (15)

a) (i) $x^3 + x + 12 = 0$

$x^3 + x = -12$
 $x^2(x^2 + 1) = -12$

$x^2 = -12$

$x = \pm 2i\sqrt{3}$

$x = \pm 2i\sqrt{3}$

(ii) as $x^2 < 0$ there must be an imaginary root.

However, $x^2 + x + 12 = 0$ is a real, roots appear in conjugate pairs.

$\therefore$ there is one real root

(iii) $P(-3) = -27 - 3 + 12 = -18 < 0$
 $P(2) = 8 - 2 + 12 = 18 > 0$

as polynomial is continuous

$\therefore$ $-3 < x < 2$

(iv) $x^2 + x = -12$
 $x^2 + x + 12 = 0$

$\therefore 4 < |x|^2 < 6$
 $2 < |x| < \sqrt{6}$

b) (i) $\frac{dx}{dt} = \frac{10x - v}{10}$
 $\frac{dx}{dt} = \frac{10x - v}{10}$

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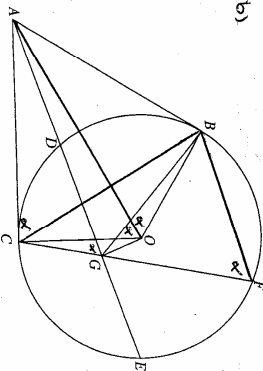
$\frac{dx}{dt} = \frac{10x - v}{10}$

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$\frac{dx}{dt} = \frac{10x - v}{10}$

b)



(i) $\angle OBA = 90^\circ$ (radius $\perp$ tangent)
 $\therefore \angle OCA = 90^\circ$

$\therefore \angle ABC$ is cyclic quadrilateral
 Opposite $\angle$ s supplementary

$\therefore \angle OCA = 90^\circ$ ($\perp$ centre bisects chord)
 $\therefore \angle OCA = \angle OCA$

$\therefore \angle OCA$ is cyclic quadrilateral
 ($\angle$ s in same segment =)

(ii) Let $\angle BAC = x$
 $\angle BOC = 2x$ (at centre twice $\angle$ at circumference on same arc)

$\angle BCA = x$ (all in circle segment)
 $\therefore \angle BOA = x$ ($\angle$ s in same segment = in $\triangle OBC$)

$\angle BOC = \angle AOB + \angle AOC$ (common $\angle$)

$\angle AOC = x$

$\angle AOC = \angle AOC$ ($\angle$ s in same segment = in $\triangle OAC$)

$\therefore \angle AOC = x$

$\therefore \angle AOC = \angle BAC = x$

$\therefore \angle AOC$ (corresponding $\angle$ s =)

c) $\frac{MN}{x} = \tan 30^\circ$

$\frac{MN}{x} = \frac{1}{\sqrt{3}}$

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Question 7) (15)

a) 

horizontal $F = m v^2$

$T \sin 60^\circ$

vertical $F = 0$

$T \cos 60^\circ = 8$

$T \cos 60^\circ + m g = 0$ (2)

$T \sin 60^\circ = \frac{m v^2}{r}$

$$\begin{aligned} \text{iii)} \quad Q(2cp, \frac{2c}{p}) \\ \frac{xy}{xy} = \frac{2cp \times \frac{2c}{p}}{4c^2} \\ \frac{xy}{xy} = \frac{4c^2}{4c^2} = 1 \end{aligned}$$

Here the result is true for $n=1$

Assume the result is true for $n=k$

$$\frac{d^{k+1}}{dx^{k+1}} (-1)^k (x \sin x - 2k \cos x)$$

Prove the result is true for $n=k+1$

$$\begin{aligned} \frac{d^{k+1}}{dx^{k+1}} &= (-1)^{k+1} (x \sin x - 2(k+1) \cos x) \\ &= (-1)^k \left[-(x \cos x) + \sin x + 2 \right] \\ &= (-1)^k [x \cos x + \sin x + 2] \\ &= (-1)^k [x(-\sin x) + (\cos x)x + 2(\sin x)] \\ &= (-1)^k [-x \sin x + (\cos x)x + 2 \sin x] \end{aligned}$$

$$\begin{aligned}
 &= -e^{-x} \cos x - \int e^{-x} \cos x \, dx \\
 &u = e^{-x} \quad v = \sin x \\
 &du = -e^{-x} dx \quad dv = \cos x \, dx
 \end{aligned}$$

$$\begin{aligned} &= \frac{1}{2} \left(e^{-(2n-1)\pi} - e^{-(2n-2i)\pi} \right) \\ &= \frac{1}{2} \left(e^{-2n\pi} + (-1-2n)\pi \right) \quad (2) \end{aligned}$$

$$= \frac{e^{\pi} + 1}{z(e^{\pi} - 1)}$$